EP

EZHIL P

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SUMMARY

Creative and passionate **Software Engineer** with **3 years** of experience in designing and implementing innovative software solutions. Proficient in **Python** and **Django framework**, with additional experience in **Golang**, **REST APIs**, and **DevOps**. Skilled in databases like **InfluxDB**, **PostgreSQL**, and **Redis**, and technologies such as **Kubernetes**, **Docker**, **AWS**, and **Jenkins**. Experienced in API testing with Postman and messaging protocols like RabbitMQ. Proven ability to enhance system performance and resolve complex technical issues while collaborating effectively to achieve project goals.

SKILLS

- **Programming Languages:** Python (Django Framework), Go
- **Data Management and Database:** InfluxDB, Flux, PostgreSQL, Redis
- **Version Control System:** GitHub, GitLab, Bitbucket
- **Cloud and Containerization:** Kubernetes, Minikube, Docker, and AWS
- **Networking and messaging protocols:** HTTP, HTTPS, MQTT, AMQP, RabbitMQ
- **Operating Systems:** Linux, Windows
- **Project Methodologies:** Agile
- **Project Management Tools:** Jira, Confluence
- **CICD and Automation Tools:** Jenkins, Celery
- **Development and Testing Tools:** Postman

EXPERIENCE SUMMARY

Software Engineer / VVDN TECHNOLOGIES PVT.LTD - Pollachi *08/2021 - Current*

- Over 3 years of experience as a Software Engineer, with expertise in tools, frameworks, and methodologies.
- Skilled in designing and developing web-based applications using **Python** and the **Django framework**, ensuring optimal performance and scalability.
- Strong analytical, debugging, and problem-solving abilities, with a proactive approach to addressing complex challenges.
- Experienced in Agile methodology, participating in daily scrums, sprint planning, customer meetings, and ensuring the timely delivery of projects.
- Proficient in working with both NoSQL (**InfluxDB**) and SQL (**PostgreSQL**) databases.
- Successfully estimated project efforts from inception to completion, adhering to deadlines, and delivering results.
- Developed and maintained **microservices** using **GoLang** and **Python**, contributing to modular and scalable architectures.
- Conducted application security testing, including vulnerability scans, port analysis, and directory exposure checks.
- Built and containerized Django-based applications with Docker for efficient deployment and scalability.
- Automated workflows using shell scripting and deployed applications on AWS EC2 instances.
- Managed AWS services, including **IAM, EC2,** and **ELB**, to ensure secure and efficient cloud operations.
- Utilized **RabbitMQ** for asynchronous communication, **Celery** for task management, and Redis for caching and in-memory data storage, optimizing application efficiency.
- Quickly adapted to evolving requirements, contributing to project agility, and alignment with user needs.

- Collaborated with cross-functional teams, including network engineers, and system administrators, to gather requirements and deliver technical solutions.
- Implemented backend components, including APIs, data models, and business logic, ensuring robust application performance.
- Created comprehensive MOPs (Methods of Procedure), outlining implementation processes and technical details.
- Partnered with QA teams to analyze and resolve software defects, enhancing stability and reliability.
- Played a key role in implementing features to track and analyze real-time device activity, improving user functionality.
- Enhanced and maintained Python code for Magma, implementing new features, and resolving critical issues.
- Conducted unit testing for development releases to validate functionality and address bugs.
- Provided detailed task updates in Jira, including relevant documentation, screenshots, and progress summaries.
- Proactively addressed complex technical challenges with cross-functional teams to achieve project goals efficiently.

PROJECTS

Project 1: Cloud Network Management System (CNMS)
Description:
The Cloud Network Management System (CNMS) is a comprehensive application suite designed to simplify network access management. It offers robust tools for monitoring, tracking, and controlling device usage and configurations, while supporting multi-level management accounts for secure and role-based access.
Technologies:
RabbitMQ | Redis | Celery | PostgreSQL | Postman | Swagger | Kubernetes | Docker
Role Description:

- Integrated **Swagger** for interactive API exploration, enhancing the user experience.
- Successfully migrated the application to production, meeting deadlines with zero disruptions.
- Deployed and debugged services on cloud platforms using **Kubernetes** and **Docker**.
- Developed APIs and applications following SDD documentation and standards.
- Created detailed API flow diagrams and MOPs (Methods of Procedure) for feature implementation.
- Conducted root cause analysis to debug and resolve issues in QA and production environments.
- Wrote efficient SQL queries to process complex data for reporting modules.
- Designed and implemented **Celery** tasks for parallel processing, optimizing resource utilization and execution time.
- Leveraged **RabbitMQ** with **MQTT protocol** for device communication through statistics, ensuring efficient data exchange and improved device management.
- Designed and developed Dockerfiles to streamline backend application deployment.

Project 2: 5G SD-WAN Project
Description:
The 5G SD-WAN Application ensures secure and seamless connectivity for 5G devices. It optimizes network performance through split tunneling and policy-based routing, ensuring quality service for enterprise applications. Integrated with **Zabbix**, it enables monitoring of device metrics and facilitates remote configuration and management.
Technologies:
RabbitMQ | Redis | Celery | PostgreSQL | Zabbix | Syslog server | Wireguard | Postman | Burp Suite | nmap | SQL Injection | OWASP | PACU | Swagger | Docker
Role Description:

- Developed APIs and applications based on SDD documentation, ensuring high-quality deliverables.
- Created parallel processing tasks using **Celery** to improve operational efficiency.
- Integrated APIs with **Zabbix** for real-time monitoring of device metrics.
- Conducted comprehensive security testing using tools like **Burp Suite**, **OWASP**, **nmap**, and **PACU**.

- Implemented **Swagger** for intuitive API documentation and interaction.
- Automated test scripts using the **Robot Framework**, ensuring reliable application testing.
- Configured AWS services, including **EC2** and **IAM**, for secure deployment and user management.
- Designed API flow diagrams and MOPs for streamlined feature implementation.
- Added backend support for an email notification system, dynamically generating emails using static HTML templates.

3: Inventory Management Project

Description:

A web-based application designed to support production units, serving as a shop floor management system. It handles essential workflows, including shipments, binning, QC, shelving, and kitting processes, while seamlessly integrating with SAP systems.

Technologies:

Python | PostgreSQL | VM | Redis | Celery | On-premises | Linux | SAP | Locker Device

Role Description:

- Collaborated with clients to gather requirements and ensure successful implementation with the development team.
- Worked onsite in production units to understand and fulfill specific operational needs.
- Integrated the web application with hardware devices, such as printers and locker devices, over Wi-Fi.
- Coordinated with the SAP team to understand workflows and designed software to align with SAP integration.

4: RAN Intelligent Controller (RIC)

Description:

The RAN Intelligent Controller (RIC) is a 5G technology designed to enhance network functionality through real-time and near-real-time control and optimization. Deployed over existing 5G networks, RIC focuses on features such as traffic steering for seamless cell handovers and energy-saving xApps to optimize power consumption across network components.

Technologies:

Python | InfluxDB | GoLang | Redis | Kubernetes

Role Description:

- Developed a **NetConf Server** used for communicating between two servers, enabling efficient configuration management.
- Dockerized services to ensure efficient deployment and management of microservices.
- Created a microservice in **GoLang** to manage alarms for RIC, ensuring timely identification and resolution of network issues.
- Conducted R&D on new 5G features and implemented them according to industry standards.
- Performed code debugging and resolved critical bugs to maintain application stability.
- Researched and developed new features in alignment with evolving 5G requirements.

EDUCATION AND TRAINING

Internship: Software Engineering *05/2022*
VVDN TECHNOLOGIES PVT Ltd - Pollachi

Bachelor of Engineering: Computer Science *04/2022*
KPR Institute of Engineering And Technology - Coimbatore